

Credit Risk Assessment

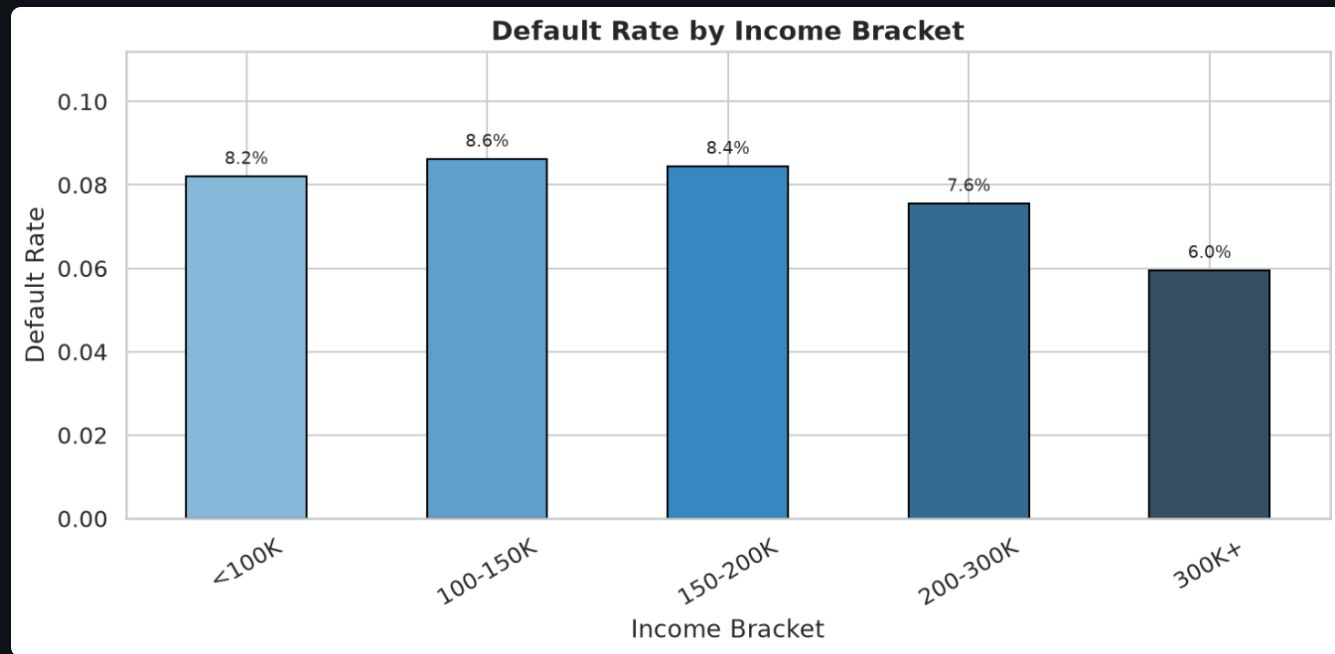
RAKSHITHA A MANOJ

Exploratory Data Analysis

Navigate

- ☒ EDA
- ☐ Risk Prediction
- ☐ Explainability
- ☐ Business Rules
- ☐ Talk to Data
- ☐ Audit Log

Default rate by income bracket

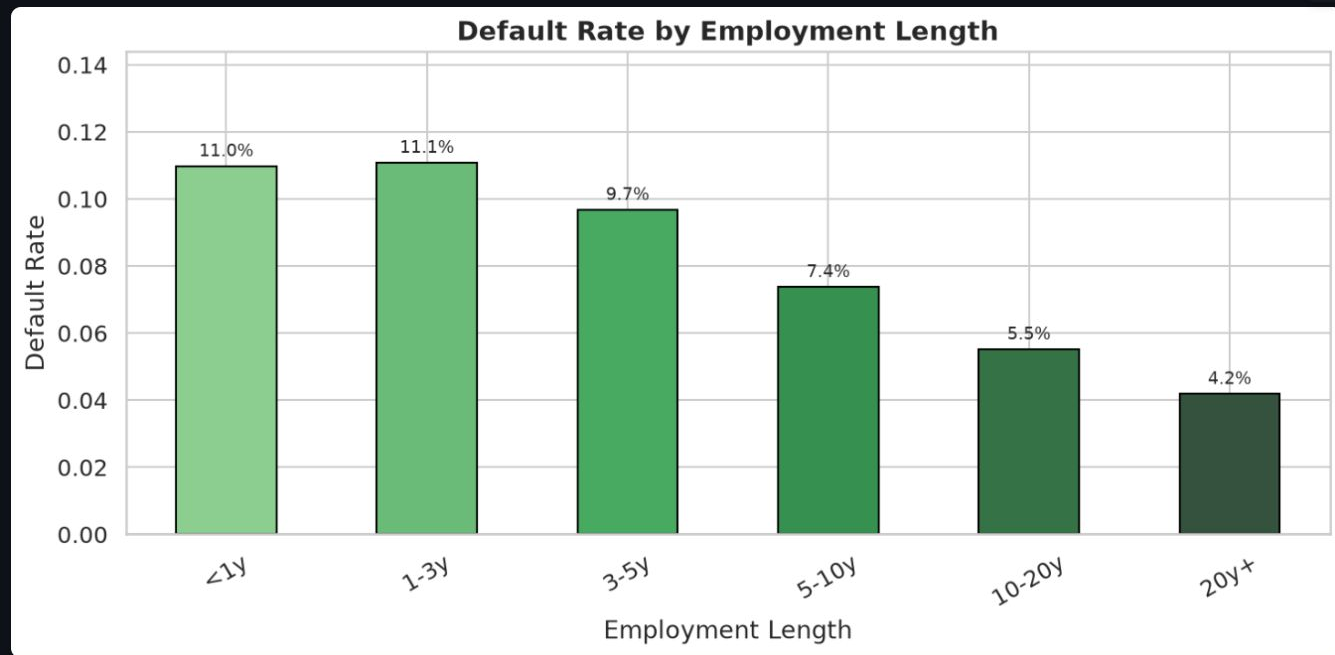


Lower-income brackets show elevated default rates; non-linear risk distribution across upper income tiers.

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Default rate by employment length

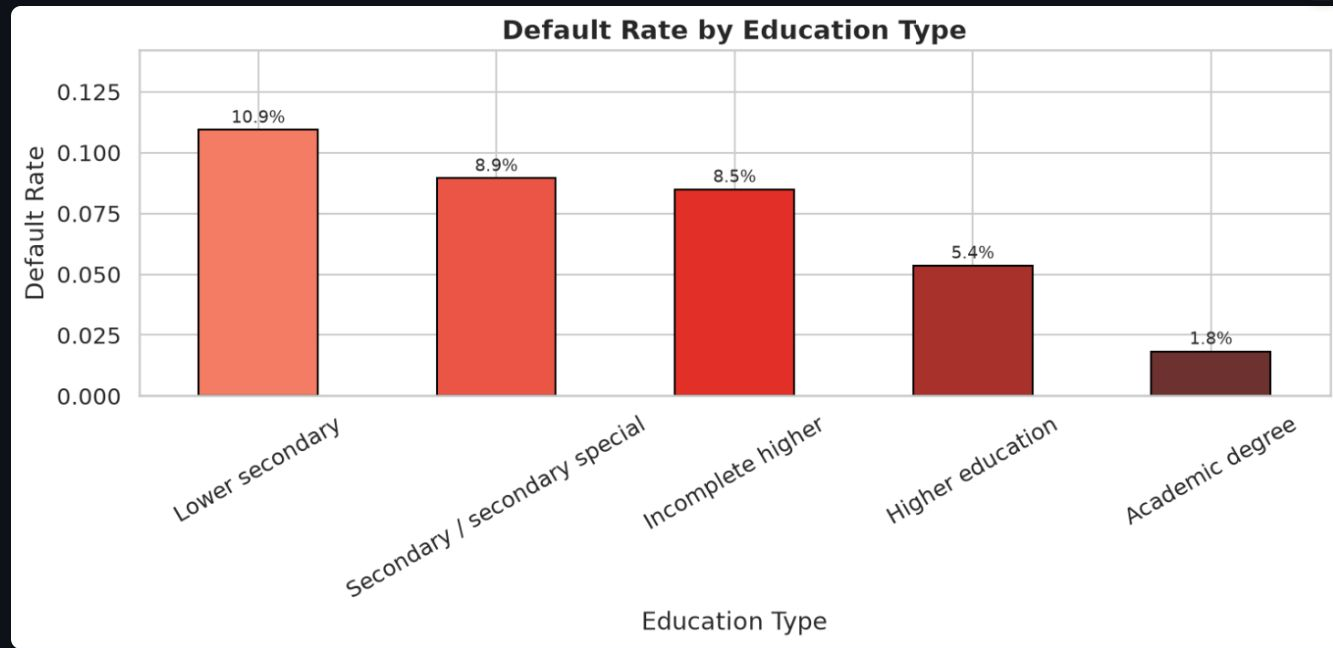


Shorter tenure correlates strongly with default risk; applicants with <1 year show highest default probability.

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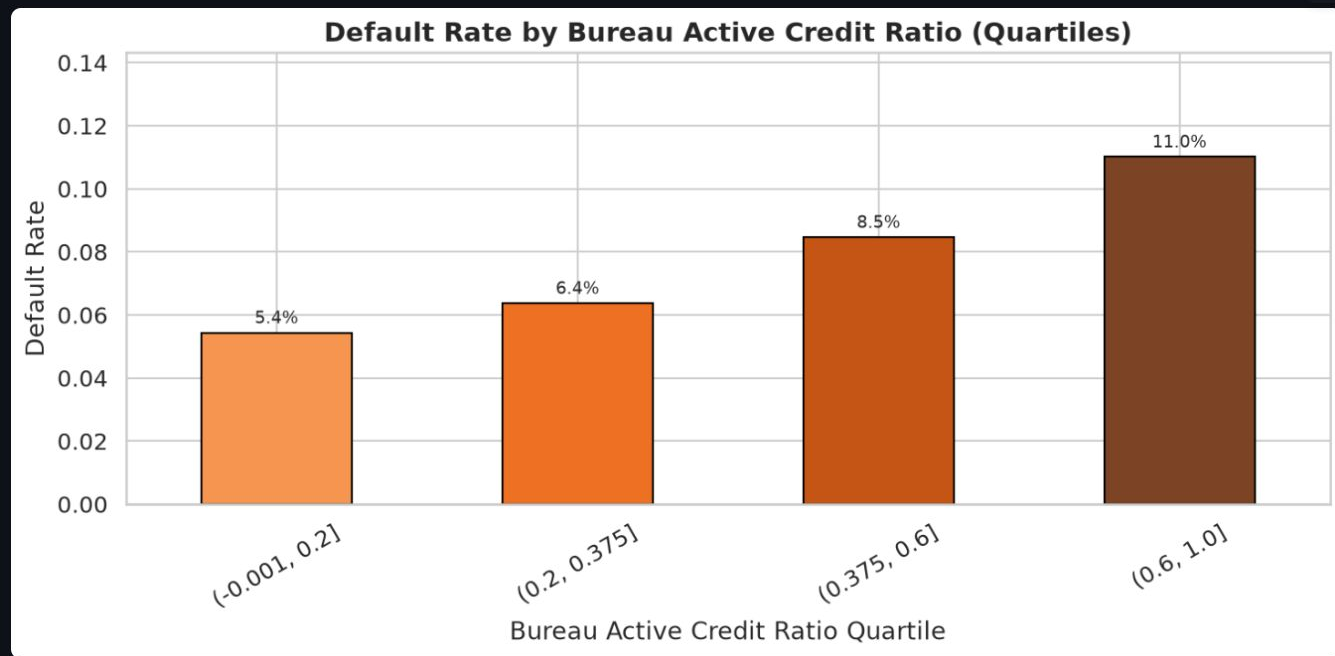
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Default rate by education type



Clear educational gradient: lower secondary exhibits highest default rates, academic degree holders lowest.

Default rate by bureau active credit ratio

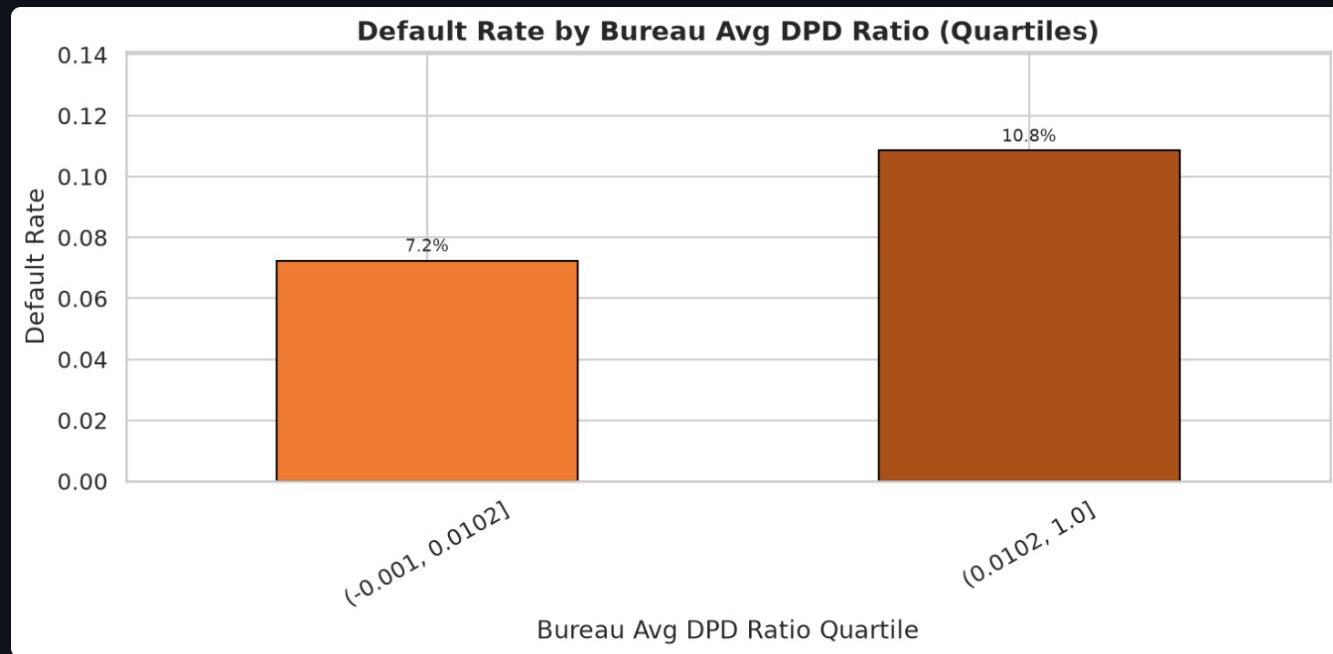


Higher proportions of actively open bureau lines indicate elevated credit utilization and risk.

Navigate

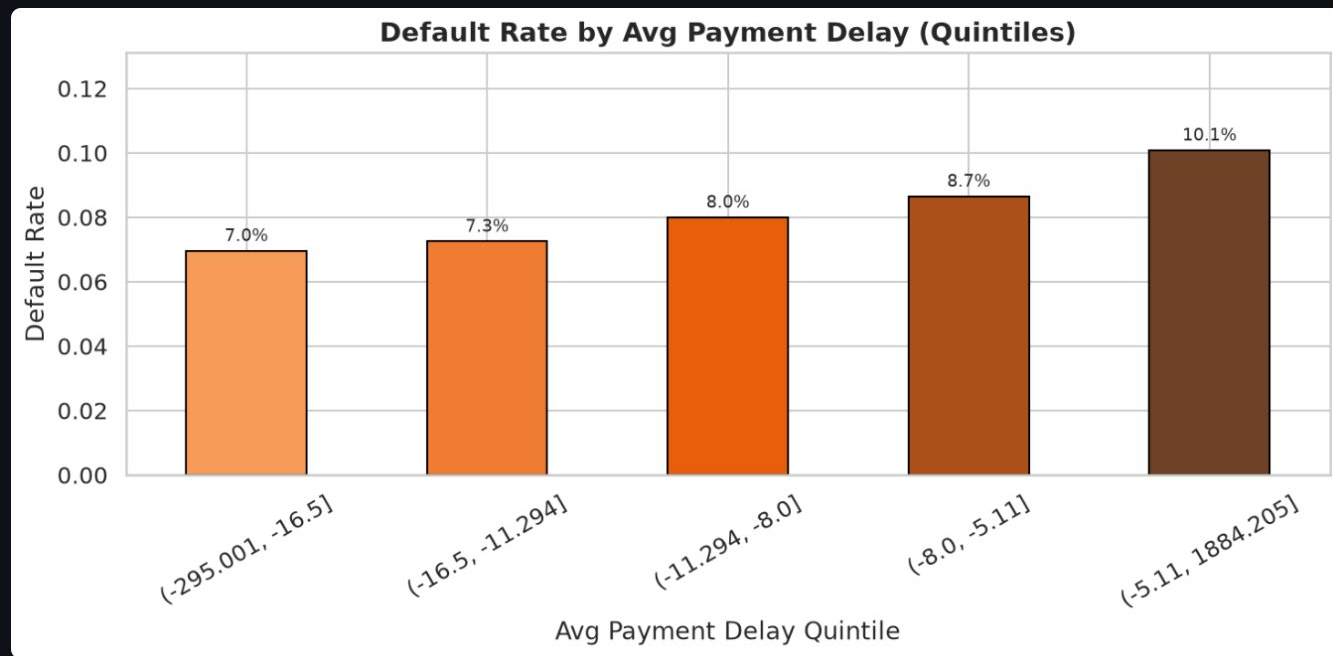
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Default rate by bureau avg DPD ratio



Historical days-past-due frequency across external bureau accounts is a powerful default predictor.

Default rate by avg payment delay

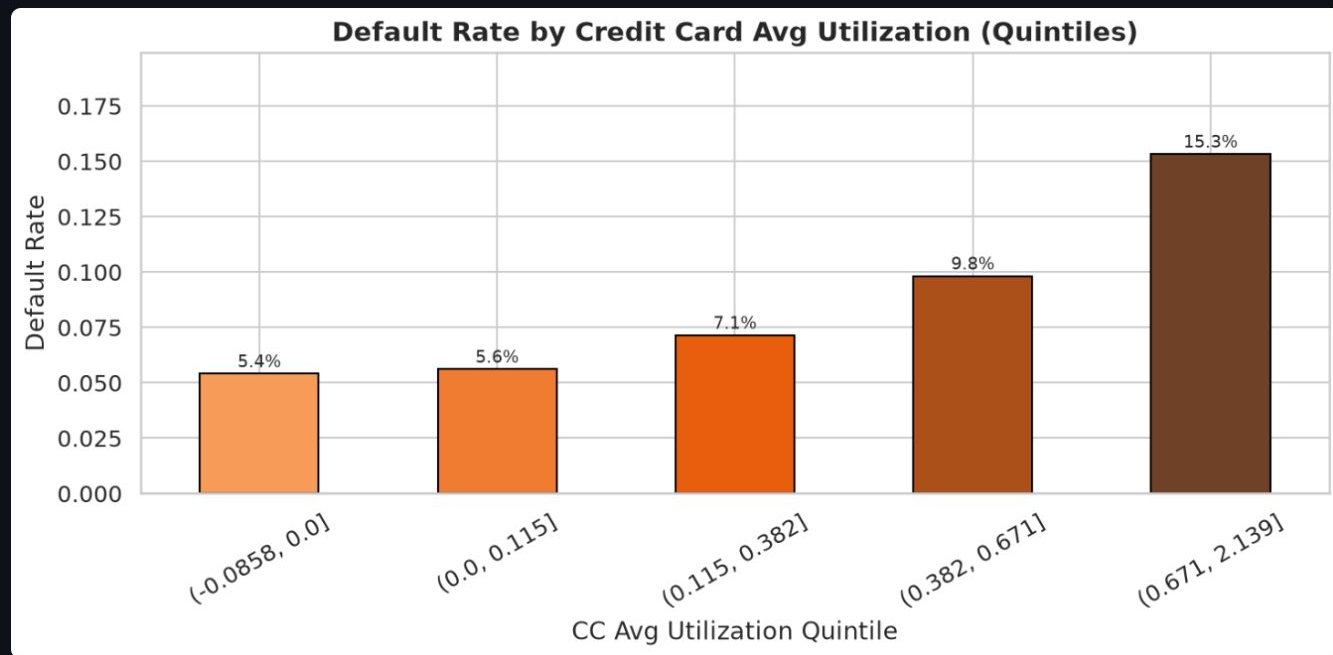


Payment delay duration on prior internal installments correlates monotonically with default.

Navigate

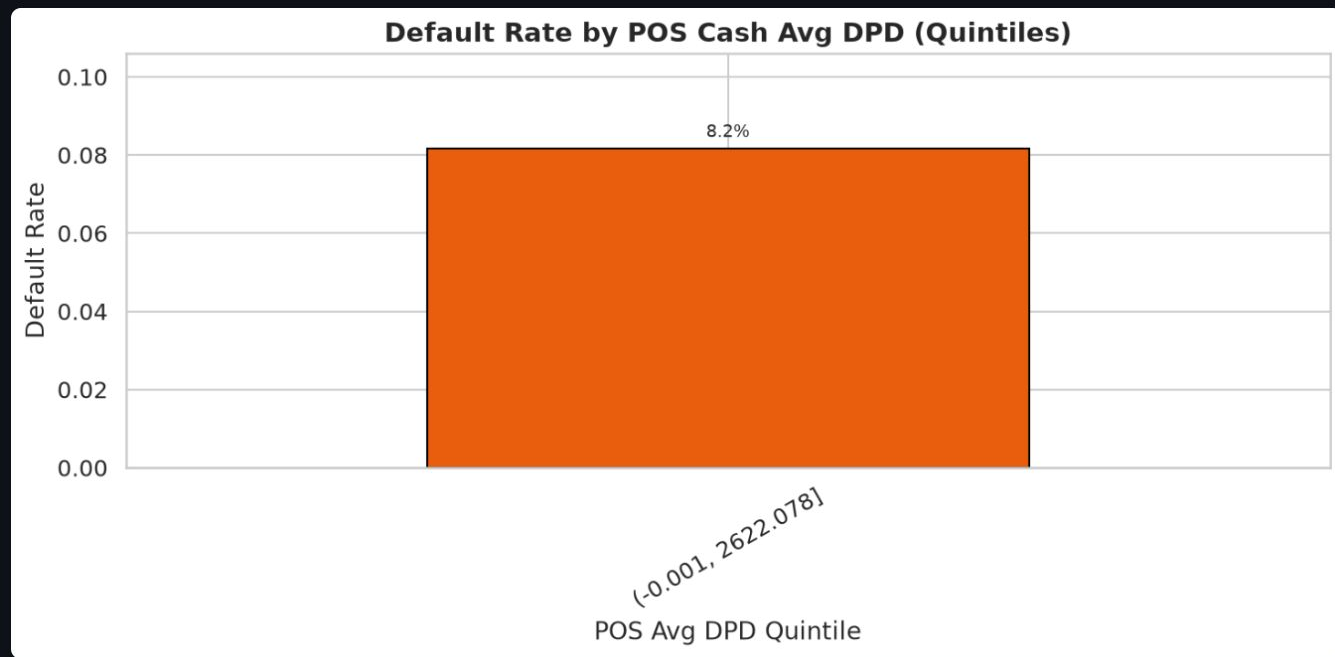
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Default rate by credit card utilization



High revolving credit card balance utilization signals liquidity strain and heightened risk.

Default rate by POS/cash loan DPD



Past delinquencies on point-of-sale and cash loans directly elevate current default probability.

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Risk Prediction

Navigate

- EDA
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Risk Prediction

Upload applicant data in CSV format to score with the calibrated LightGBM model.

Upload applicant data (CSV)

 test_sample_20.csv

12.9KB

Successfully scored 20 applicants.

Prediction Results

	SK_ID_CURR	default_probability	risk_band
0	384575	0.0766	Medium
1	214010	0.0191	Low
2	142232	0.0782	Medium
3	389171	0.0117	Low
4	283617	0.0512	Medium
5	362171	0.1422	Medium
6	180689	0.0234	Low

Navigate

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16	231383	0.004	Low
17	119647	0.1256	Medium
18	257374	0.0558	Medium
19	305211	0.0149	Low

Risk Band Distribution



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Deploy

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Risk Prediction

Explainability

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AI-Powered Credit Risk Intelligence Platform

Risk Prediction

Upload applicant data in CSV format to score with the calibrated LightGBM model.

Upload applicant data (CSV)

test_single_row.csv

2.5KB

+

Successfully scored 1 applicants.

Prediction Results

	SK_ID_CURR	default_probability	risk_band
0	174545	0.0187	Low

Risk Band Distribution

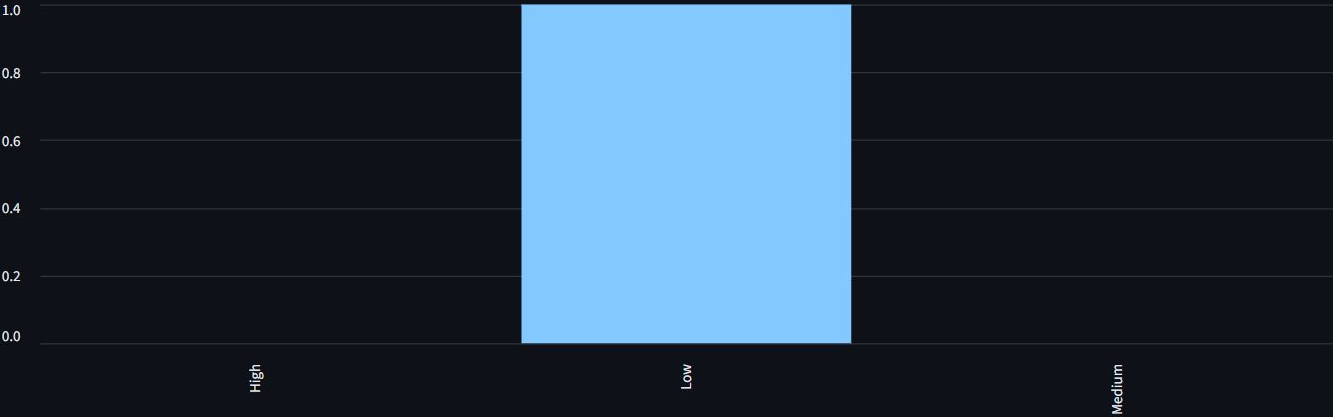
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Prediction Results

	SK_ID_CURR	default_probability	risk_band
0	174545	0.0187	Low

Risk Band Distribution



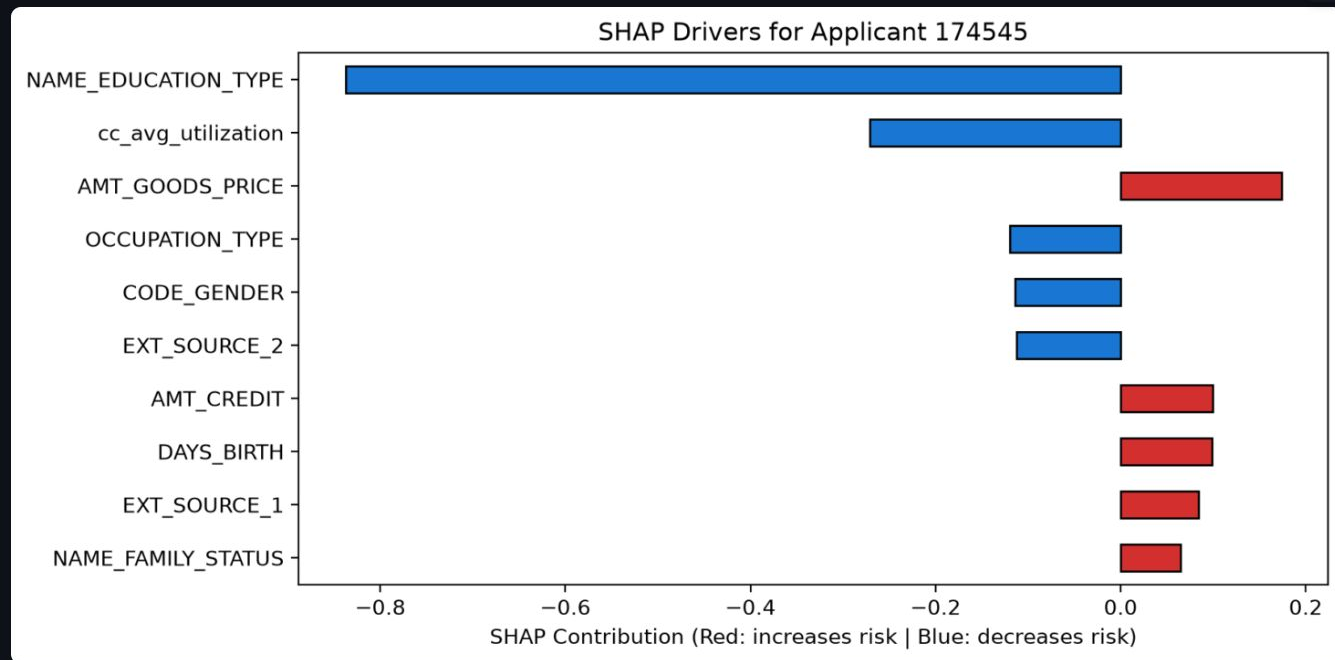
Explainability

Deploy

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Top 10 Feature Contributions (SHAP)

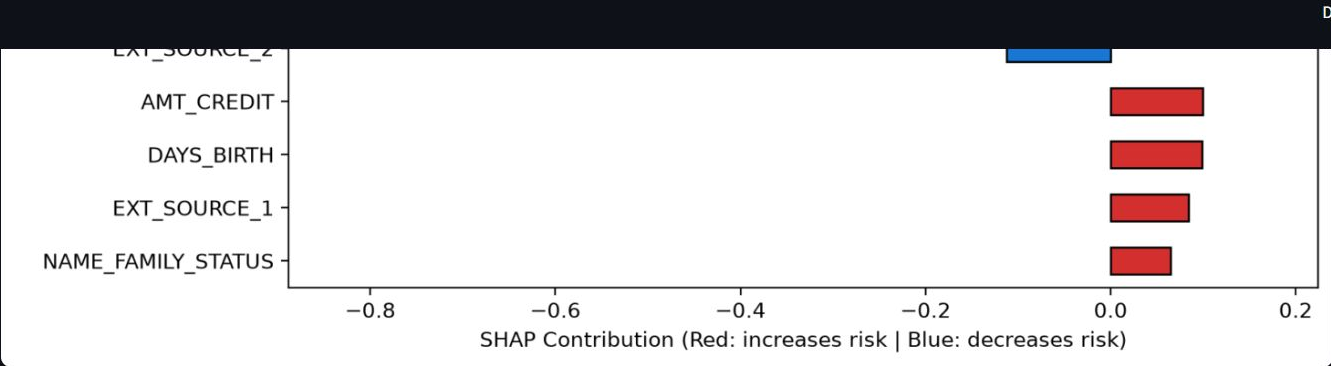


Calibrated Default Probability

Assigned Risk Band

Navigate

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Calibrated Default Probability

1.87%

Assigned Risk Band

Low

AI Analysis

Their education type, low average credit-card utilization, stable occupation type, and gender code all decrease the chance of default. However, the relatively high price of the goods they intend to purchase increases the default risk.

Business Rules

Navigate

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Derived Business Rules

Transparent surrogate decision-tree rules and descriptive feature drivers.

Surrogate Fidelity

89.2%

Fidelity measures how closely the surrogate decision tree approximates the LightGBM base model on a 500-row random sample (drawn from 307,511 training records). Values of 85–95% indicate a reliable policy approximation; 100% on small samples may indicate overfitting.

Surrogate Decision-Tree Rules

```
|--- EXT_SOURCE_3 <= 0.54
|   |--- EXT_SOURCE_2 <= 0.54
|   |   |--- cc_avg_utilization <= 8193.50
|   |   |   |--- BASEMENTAREA_AVG <= 0.08
|   |   |   |   |--- class: 0
|   |   |   |   |--- BASEMENTAREA_AVG > 0.08
|   |   |   |   |--- class: 1
|   |   |--- cc_avg_utilization > 8193.50
|   |   |   |--- DAYS_BIRTH <= -19848.50
|   |   |   |   |--- class: 1
|   |   |   |   |--- DAYS_BIRTH > -19848.50
|   |   |   |   |   |--- class: 1
|   |   |--- EXT_SOURCE_2 > 0.54
```

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| | | |--- class: 0

| | | |--- prev_app_count > 2.00

| | | |--- class: 1

| | | |--- EXT_SOURCE_2 > 0.11

| | | |--- bureau_avg_days_credit <= -460.50

| | | |--- pos_avg_dpd <= 0.44

| | | | |--- class: 0

| | | | |--- pos_avg_dpd > 0.44

| | | | |--- class: 0

| | | |--- bureau_avg_days_credit > -460.50

| | | |--- DAYS_BIRTH <= -17518.50

| | | | |--- class: 0

| | | |--- DAYS_BIRTH > -17518.50

| | | | |--- class: 1

Descriptive SHAP Summary

• Applicants with EXT_SOURCE_2 deviating from the typical value (median: 0.55) are a significant driver of risk classification.

• Applicants with EXT_SOURCE_3 deviating from the typical value (median: 0.54) are a significant driver of risk classification.

• Applicants with EXT_SOURCE_1 deviating from the typical value (median: 0.51) are a significant driver of risk classification.

• Applicants with AMT_CREDIT deviating from the typical value (median: 510772.50) are a significant driver of risk classification.

• Applicants with CODE_GENDER deviating from the typical value (median: 0.00) are a significant driver of risk classification.

Talk To Data

Navigate

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Deploy ⋮

Ask a question about the applicant data:

What is the average credit amount for Low risk applicants?

Run Query

Answer

The data show that low-risk applicants have an average credit amount of roughly \$638,057. This figure comes directly from the credit-amount field in the dataset.

▼ Show generated SQL

```
SELECT AVG(AMT_CREDIT) FROM applicants WHERE risk_band = 'Low';
```

Query Result

	AVG(AMT_CREDIT)
0	638057.346

Navigate

- ☐ EDA
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Ask a question about the applicant data:

Summarize the profile of a typical High risk applicant

Run Query

Answer

Typical profile of a “High-risk” applicant (based on the 49,970 records in the data set)

Characteristic	Average value
Total annual income	≈ 158,732 (currency units)
Total credit amount (all loans/lines)	≈ 511,308
Monthly annuity payment	≈ 25,862
Length of employment (days)	≈ 24,819 days (about 68 years)
Model-estimated probability of default	≈ 23.9 %
Ratio of active bureau-reported credit to total credit	≈ 56.1 %
Ratio of previously approved applications to total applications	≈ 65.7 %
Ratio of completed point-of-sale (POS) transactions to total POS attempts	≈ 8.3 %
Average credit-card utilization (balance ÷ limit)	≈ 56.8 %

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- All figures are simple averages across the high-risk group; individual applicants may differ markedly from these central tendencies.

Show generated SQL

```
SELECT
  COUNT(*) AS applicant_count,
  AVG(AMT_INCOME_TOTAL) AS avg_income,
  AVG(AMT_CREDIT) AS avg_credit,
  AVG(AMT_ANNUITY) AS avg_annuity,
  AVG(DAYS_EMPLOYED) AS avg_days_employed,
  AVG(default_probability) AS avg_default_probability,
  AVG(bureau_credit_active_ratio) AS avg_bureau_credit_active_ratio,
  AVG(prev_approved_ratio) AS avg_prev_approved_ratio,
  AVG(pos_completed_ratio) AS avg_pos_completed_ratio,
  AVG(cc_avg_utilization) AS avg_cc_utilization
FROM applicants
WHERE risk_band = 'High';
```

Query Result

	applicant_count	avg_income	avg_credit	avg_annuity	avg_days_employed	avg_default_probability	avg_bureau_credit_active_ratio	avg_prev_approved_r
0	49970	158732.1087	511308.4492	25862.1062	24818.9594	0.2393	0.5606	0.6

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Ask a question about the applicant data:

what's the average debt-to-income ratio

Run Query

Answer

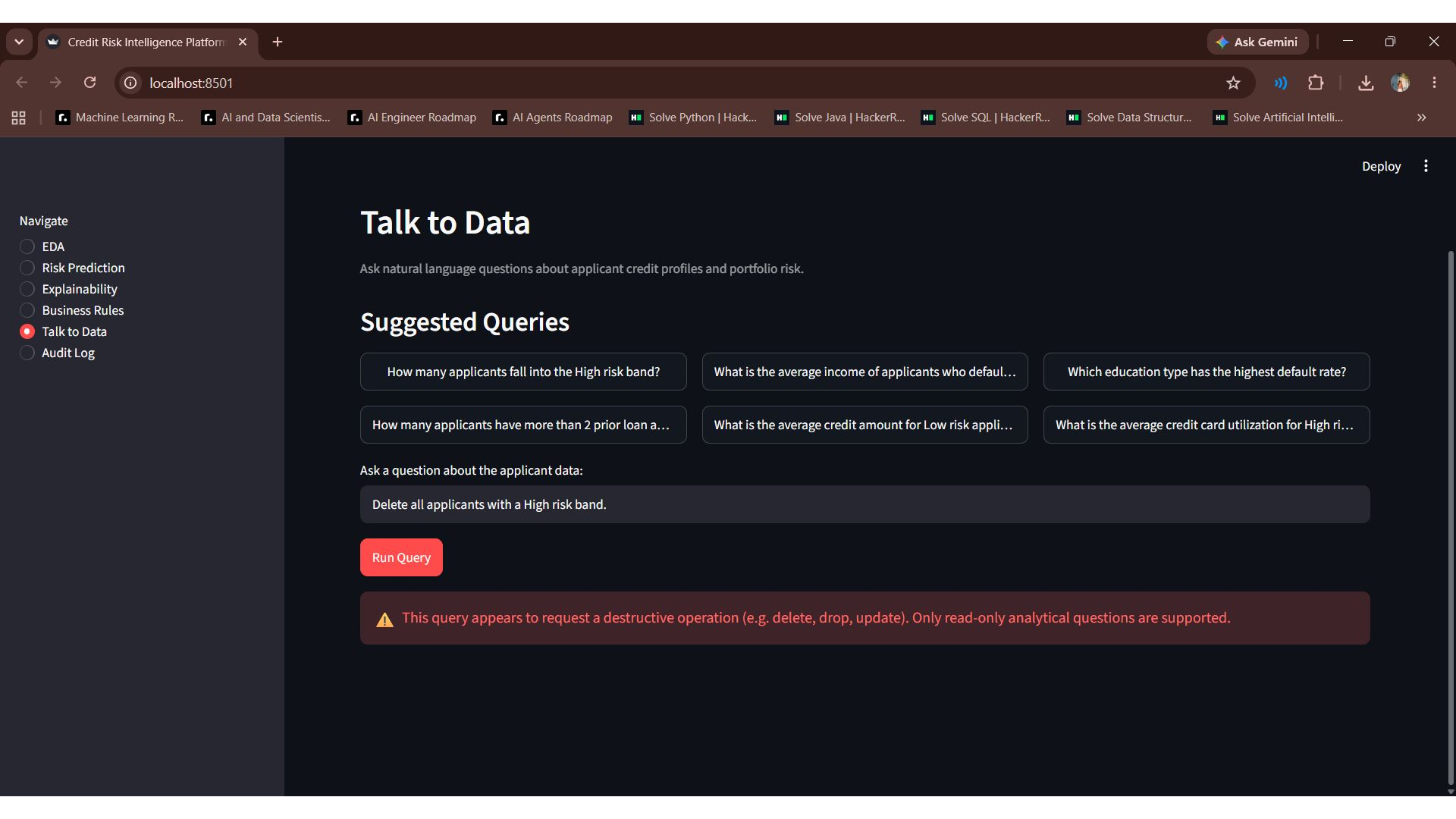
The dataset does not contain a standard debt-to-income (DTI) ratio. The query therefore used a rough proxy—total credit amount divided by total annual income (AMT_CREDIT / AMT_INCOME_TOTAL). Note that the industry-standard DTI is usually calculated as **monthly debt service ÷ monthly gross income**, so this proxy is not directly comparable.

Using that proxy, the average across all applicants is **approximately 3.96** (i.e., total credit is about 4 times the applicant's annual income).

> Show generated SQL

Query Result

	avg_debt_to_income_ratio
0	3.9576



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Talk to Data

Ask natural language questions about applicant credit profiles and portfolio risk.

Suggested Queries

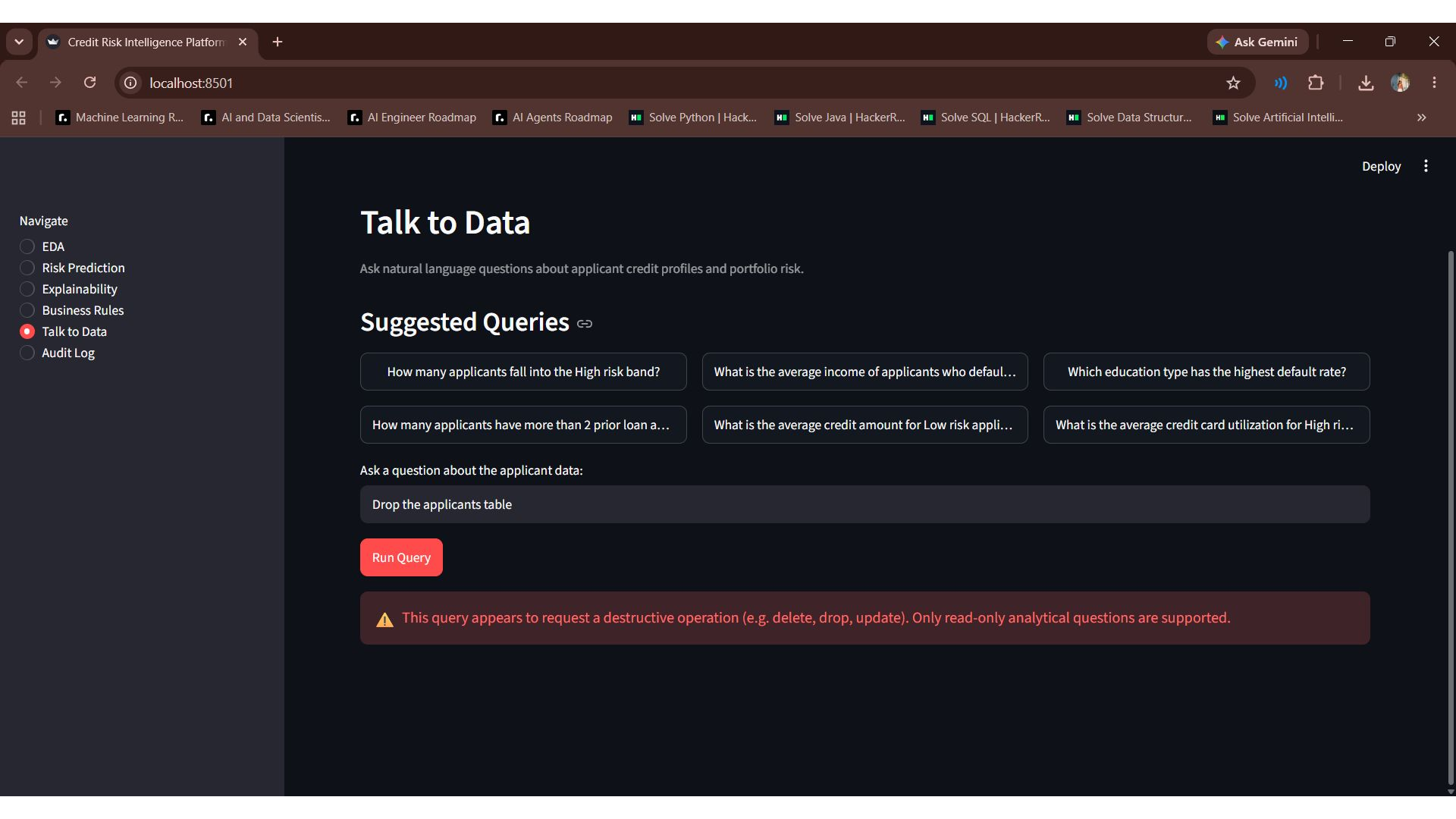
- How many applicants fall into the High risk band?
- What is the average income of applicants who default...
- Which education type has the highest default rate?
- How many applicants have more than 2 prior loan a...
- What is the average credit amount for Low risk appli...
- What is the average credit card utilization for High ri...

Ask a question about the applicant data:

Delete all applicants with a High risk band.

Run Query

⚠️ This query appears to request a destructive operation (e.g. delete, drop, update). Only read-only analytical questions are supported.



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Suggested Queries

How many applicants fall into the High risk band?

What is the average income of applicants who default...

Which education type has the highest default rate?

How many applicants have more than 2 prior loan a...

What is the average credit amount for Low risk appli...

What is the average credit card utilization for High ri...

Ask a question about the applicant data:

Drop the applicants table

Run Query

⚠️ This query appears to request a destructive operation (e.g. delete, drop, update). Only read-only analytical questions are supported.

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Which education type has the highest default rate?

Run Query

Answer

Based on the data, the education group with the highest observed default rate is “**Lower secondary**”, where about **10.9%** of applicants defaulted.

The next highest rates are:

- **Secondary / secondary special** – 8.9%
- **Incomplete higher** – 8.5%
- **Higher education** – 5.4%
- **Academic degree** – 1.8%

These percentages come from averaging the binary default indicator (1=default, 0=no default) for each education category in the dataset. No other education types in the data exceed the 10.9% default rate observed for “Lower secondary.”

▼ Show generated SQL

```
SELECT NAME_EDUCATION_TYPE, AVG(TARGET) AS default_rate
FROM applicants
GROUP BY NAME_EDUCATION_TYPE
ORDER BY default_rate DESC
LIMIT 25;
```

Audit Trail

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Audit Trail

Immutable record of scored applicants and natural language chatbot exchanges.

PredictionsChat History

	id	timestamp	applicant_id	default_probability	risk_band	top_shap_drivers
0	160	2026-09-06T17:02:16.043375+00:00	174545	0.0187	Low	None
1	159	2026-09-06T17:01:48.027904+00:00	305211	0.0149	Low	None
2	158	2026-09-06T17:01:47.994959+00:00	257374	0.0558	Medium	None
3	157	2026-09-06T17:01:47.964559+00:00	119647	0.1256	Medium	None
4	156	2026-09-06T17:01:47.934354+00:00	231383	0.004	Low	None
5	155	2026-09-06T17:01:47.900029+00:00	348749	0.1807	High	None
6	154	2026-09-06T17:01:47.855653+00:00	250804	0.1156	Medium	None
7	153	2026-09-06T17:01:47.798428+00:00	347275	0.0349	Low	None
8	152	2026-09-06T17:01:47.759647+00:00	188909	0.0694	Medium	None
9	151	2026-09-06T17:01:47.717382+00:00	120952	0.1079	Medium	None

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Predictions

Chat History

	id	timestamp	question	generated_sql
0	17	2026-09-06T17:09:37.626608+00:00	Which education type has the highest default rate?	SELECT NAME_EDUCATION_TYPE, AVG(TARGET) AS defau
1	16	2026-09-06T17:07:53.731477+00:00	Summarize the profile of a typical High risk applicant	SELECT COUNT(*) AS applicant_count, AVG(AMT_INC
2	15	2026-09-06T17:06:36.819257+00:00	What is the average credit amount for Low risk applicants?	SELECT AVG(AMT_CREDIT) FROM applicants WHERE risk_
3	14	2026-09-06T16:39:13.008955+00:00	Summarize the profile of a typical High risk applicant	SELECT AVG(AMT_INCOME_TOTAL) AS avg_income, A
4	13	2026-09-06T16:38:18.022110+00:00	Which applicants look the riskiest overall?	SELECT SK_ID_CURR, default_probability, risk_band FRC
5	12	2026-09-06T16:34:17.967641+00:00	What's the average debt-to-income ratio?	SELECT AVG(AMT_CREDIT * 1.0 / AMT_INCOME_TOTAL) A
6	11	2026-09-06T16:32:32.288540+00:00	What is the average FICO score for each risk band?	SELECT risk_band, AVG(default_probability) AS avg_d
7	10	2026-09-06T16:25:21.673074+00:00	How many applicants have a bureau credit active ratio above 50%?	SELECT COUNT(*) FROM applicants WHERE bureau_credi
8	9	2026-09-06T16:24:39.817630+00:00	What is the average annuity amount for High risk applicants?	SELECT AVG(AMT_ANNUITY) FROM applicants WHERE ris
9	8	2026-09-06T16:23:03.676945+00:00	Which education type has the highest default rate?	SELECT NAME_EDUCATION_TYPE FROM applicants GROU